

SCS — Internal Technical Spec

Score-only sparse-electrode attack on EEG recognition

Internal. Edit this first; the manuscript follows it.

1 Question

Question. Under *score-only* access (the adversary sees class logits, not gradients or weights), how few *complete electrode traces* must be controlled to flip an EEG subject-recognition decision using only EEG-band waveforms, and *what quantity governs that number*? The question is open because dense, white-box, and channel-time attacks do not measure whole-channel cost, and because the *mechanism* behind a small count — not just its value — is unknown.

2 Setup and notation

All objects are typed before use. The trial index is carried on every per-trial object; it is dropped only for *operators* whose value does not depend on which trial they act on (the margin and the waveform map).

Notation typography rule. Scalars use ordinary italic or Greek letters, e.g. $c, k, t, \rho, \Delta, \nu$ and the declared scalar scores $F_c, \hat{F}_c$. Vectors use bold lowercase, e.g. $\mathbf{u}_j$ and $\mathbf{s}(\mathbf{E})$. Matrices or multi-channel trials use bold uppercase, e.g. $\mathbf{X}_i, \mathbf{E}_i$, and $\mathbf{A}_i^{(k)}$. Sets use calligraphic uppercase, e.g. $\mathcal{D}, \mathcal{C}, \mathcal{B}$, and $\mathcal{S}^{(k)}$. In particular, $s_c(\mathbf{E})$ is the scalar channel indicator, while $\mathcal{S}(\mathbf{E})$ is the corresponding channel set; $e_{i,c}^{(k)}(t)$ is a scalar waveform value, while $\mathbf{E}_{\mathcal{S}^{(k)}, \mathbf{A}_i^{(k)}}$ is the full perturbation matrix.

Indexing and dependency invariant. Before any equation is accepted, every symbol must declare what it depends on, its size/shape, and the set or source it comes from. Indexing is one case of this rule: if the right-hand side depends on trial i , channel c , atom m , support size k , probe draw j , probe sample q , or optimization step ℓ , then the left-hand side must expose that dependency. Table 1 is the gate used for SCS notation.

Data and model. The dataset is $\mathcal{D} = \{(\mathbf{X}_i, y_i)\}_{i=1}^N$ with trials $\mathbf{X}_i \in \mathbb{R}^{C \times T}$, where C is the number of electrode channels and T is the number of samples. Let $\mathcal{C} := \{1, \dots, C\}$ be the channel index set. Labels are $y_i \in [K] := \{1, \dots, K\}$, and $i \in [N]$. The recognizer is a score map $f : \mathbb{R}^{C \times T} \rightarrow \mathbb{R}^K$ with prediction $\hat{y}(\mathbf{X}) = \arg \max_{r \in [K]} f_r(\mathbf{X})$, using a fixed deterministic tie-break. The clean predicted identity of trial i is $y_0^i = \hat{y}(\mathbf{X}_i)$.

Data-use accounting. The data are not one undifferentiated pool. The model-training split $\mathcal{D}_{\text{train}}$ is used only to fit the recognizer f . The validation split $\mathcal{D}_{\text{val}}$ is used to report clean accuracy/EER and to form the clean-correct pool $\mathcal{D}_{\text{cc}} = \{(\mathbf{X}, y) \in \mathcal{D}_{\text{val}} : \hat{y}(\mathbf{X}) = y\}$. Stage 1 uses a separate probe set $\mathcal{D}_{\text{probe}} = \{\mathbf{Z}_q\}_{q=1}^n$ to estimate channel leverage. Stage 2 evaluates attacks on

$\mathcal{D}_{\text{attack}} \subseteq \mathcal{D}_{\text{cc}}$. The final experiment must report both $n = |\mathcal{D}_{\text{probe}}|$ and $|\mathcal{D}_{\text{attack}}|$; ASR and k^* are computed only over $\mathcal{D}_{\text{attack}}$, not over all validation windows.

Object	Current HBN 122-subject artifact	Use
$\mathcal{D}_{\text{train}}$	785 passive-task recordings; 55,665 windows	train EEGConformer subject recognizer
$\mathcal{D}_{\text{val}}$	487 active-task recordings; 44,694 windows	report clean validation accuracy and biometric EER
$\mathcal{D}_{\text{cc}}$	35,042 clean-correct validation windows	candidate pool from which attack trials are drawn
$\mathcal{D}_{\text{attack}}$	122 clean-correct windows, capped by <code>attack_max_samples=122</code>	report ASR, attacked accuracy, and first-success channel count k^*
$\mathcal{D}_{\text{probe}}$	not yet regenerated for the population Stage 1 artifact	estimate F_c and $\hat{F}_c$ for the shared channel ranking

Margin (operator). For any $\mathbf{X} \in \mathbb{R}^{C \times T}$, $y \in [K]$,

$$M(\mathbf{X}, y) = f_y(\mathbf{X}) - \max_{r \neq y} f_r(\mathbf{X}). \quad (1)$$

The clean decision on trial i is correct iff $M(\mathbf{X}_i, y_0^i) > 0$; a non-targeted attack on trial i succeeds once $M(\mathbf{X}_i + \mathbf{E}_i, y_0^i) < 0$.

Perturbation, support, budget. The perturbation chosen for trial i is $\mathbf{E}_i \in \mathbb{R}^{C \times T}$ (a per-trial solution, hence indexed). The channel-support indicator is

$$s_c(\mathbf{E}) = \mathbf{1}\{\|\mathbf{E}_{c,:}\|_2 > 0\}, \quad c \in \mathcal{C},$$

with vector $\mathbf{s}(\mathbf{E}) \in \{0, 1\}^{\mathcal{C}}$. The activated channel set, when needed, is $\mathcal{S}(\mathbf{E}) = \{c \in \mathcal{C} : s_c(\mathbf{E}) = 1\}$, and the channel-count cost is $\sum_{c \in \mathcal{C}} s_c(\mathbf{E})$. The support is not fixed in advance — $\mathcal{S}(\mathbf{E}_i)$ is whatever channels the solution activates. Each perturbation obeys the peak-ratio cap $\|\mathbf{E}_i\|_\infty \leq r_{\text{max}} \|\mathbf{X}_i\|_\infty$, $r_{\text{max}} \in (0, 1)$.

Waveform basis. Before either stage runs, we build a fixed waveform dictionary $\mathcal{B} = \{b_m\}_{m=1}^r$ for one EEG window. It is not learned from the model and not resampled during the attack. It is constructed from known signal-processing choices: the window length T , sampling rate f_s , Hanning taper $h(t)$, an EEG-band frequency grid, a grid of bump centers, and low-order trend degrees. Each atom $b_m : \{0, \dots, T-1\} \rightarrow \mathbb{R}$ is normalized to unit norm, $b_m = \tilde{b}_m / \|\tilde{b}_m\|_2$. The atom indices are partitioned into three families, $\mathcal{M}_{\text{osc}} \cup \mathcal{M}_{\text{hat}} \cup \mathcal{M}_{\text{trend}} = [r]$, with sizes $r = r_{\text{osc}} + r_{\text{hat}} + r_{\text{trend}}$. For oscillatory atoms, use a fixed product grid of frequencies and phases: $\mathcal{F} = \{f_a^{\text{osc}}\}_{a=1}^{R_f} \subseteq [f_{\text{min}}, f_{\text{max}}]$ and $\mathcal{G}_\varphi = \{\varphi_b\}_{b=1}^{R_\varphi} \subset S^1$. Each $m \in \mathcal{M}_{\text{osc}}$ corresponds to one pair $(\alpha(m), \beta(m)) \in [R_f] \times [R_\varphi]$. These are fixed dictionary coordinates, not random variables sampled during the attack:

$$(\text{osc}) \quad \tilde{b}_m(t) = h(t) \cos(2\pi f_{\alpha(m)}^{\text{osc}} t / f_s + \varphi_{\beta(m)}), \quad m \in \mathcal{M}_{\text{osc}}; \quad (2)$$

$$(\text{hat}) \quad \tilde{b}_m(t) = h(t) \max(1 - |t/(T-1) - \mu_m|/w, 0), \quad m \in \mathcal{M}_{\text{hat}}; \quad (3)$$

$$(\text{trend}) \quad \tilde{b}_m(t) = h(t) (s(t)^{d_m} - \overline{s^{d_m}}), \quad s(t) = \frac{2t}{T-1} - 1, \quad m \in \mathcal{M}_{\text{trend}}. \quad (4)$$

For an active channel set $\mathcal{S} \subseteq \mathcal{C}$ and coefficient matrix $\mathbf{A} \in \mathbb{R}^{C \times r}$ with one row per global EEG channel, the induced perturbation operator uses the set indicator directly:

$$(\mathbf{E}_{\mathcal{S}, \mathbf{A}})_{c,t} = \mathbf{1}\{c \in \mathcal{S}\} \sum_{m=1}^r a_{c,m} b_m(t). \quad (5)$$

Here $\mathbf{A}$ is an operator argument. We keep it as a $C \times r$ matrix instead of a local $|\mathcal{S}| \times r$ matrix so row c always means channel c . In Stage 2, the optimized coefficient matrix is trial-and-budget dependent, written $\mathbf{A}_i^{(k)}$, so the coefficient on trial i , channel c , atom m , and budget k is $a_{i,c,m}^{(k)}$.

Sub-Gaussian. A real random variable Y with mean μ is ν -sub-Gaussian if $\mathbb{E} e^{\lambda(Y-\mu)} \leq e^{\lambda^2 \nu^2 / 2}$ for all $\lambda \in \mathbb{R}$; then $\Pr(|Y - \mu| \geq \varepsilon) \leq 2e^{-\varepsilon^2 / (2\nu^2)}$, and the mean of n i.i.d. such variables is $(\nu/\sqrt{n})$ -sub-Gaussian.

3 Threat model objective

For each trial the ideal attack is the per-trial combinatorial program

$$\min_{\mathbf{E}_i \in \mathbb{R}^{C \times T}} \sum_{c \in \mathcal{C}} s_c(\mathbf{E}_i) \quad \text{s.t.} \quad M(\mathbf{X}_i + \mathbf{E}_i, y_0^i) < 0, \quad \|\mathbf{E}_i\|_\infty \leq r_{\max} \|\mathbf{X}_i\|_\infty. \quad (6)$$

The minimizer $\mathbf{E}_i^*$ and its activated channel set $\mathcal{S}(\mathbf{E}_i^*)$ depend on $\mathbf{X}_i$. SCS is a black-box heuristic for (6), not a certificate of its optimum.

4 Method: two-stage SCS

4.1 Stage 1: population channel ranking

Let P_{probe} denote the distribution of probe trials available to the score-only attacker. Fix a finite probe set $\mathcal{D}_{\text{probe}} = \{\mathbf{Z}_q\}_{q=1}^n$ with $\mathbf{Z}_q \stackrel{\text{i.i.d.}}{\sim} P_{\text{probe}}$. The set $\mathcal{D}_{\text{probe}}$ is the sample used to estimate the ranking; it is not the distribution itself. Also fix a bank $\mathcal{P} = \{\mathbf{u}_1, \dots, \mathbf{u}_J\}$ of J sign vectors $\mathbf{u}_j \in \{-1, +1\}^r$. The bank is sampled once, e.g. with independent Rademacher entries, and then held fixed for all channels and probe trials.

For channel c and probe direction $\mathbf{u}_j$, the unscaled single-channel probe is

$$(\tilde{\mathbf{E}}_{c,j})_{c',t} = \begin{cases} \rho \sum_{m=1}^r u_{j,m} b_m(t), & c' = c, \\ 0, & c' \neq c, \end{cases}$$

where $\rho > 0$ controls the probe amplitude. For any probe-distribution trial $\mathbf{X}$, define the capped probe $\mathbf{E}_{\mathbf{X},c,j}^{\text{probe}} = \text{Cap}_{\mathbf{X}}(\tilde{\mathbf{E}}_{c,j})$, a deterministic rescaling to satisfy the peak-ratio cap. For the finite sample $\mathbf{Z}_q$, this becomes $\mathbf{E}_{\mathbf{Z}_q,c,j}^{\text{probe}}$. Thus the basis $\{b_m\}$, channel c , and amplitude ρ are fixed; the waveform randomness comes from the sampled sign vector $\mathbf{u}_j$, and the cap can make the final amplitude depend on the probe trial being evaluated.

Definition 1 (leverage). *The leverage of channel c on any probe-distribution trial $\mathbf{X}$ is the average single-channel margin drop over the probe bank,*

$$g_c(\mathbf{X}) = M(\mathbf{X}, \hat{y}(\mathbf{X})) - \frac{1}{J} \sum_{j=1}^J M(\mathbf{X} + \mathbf{E}_{\mathbf{X},c,j}^{\text{probe}}, \hat{y}(\mathbf{X})).$$

It is an average (a sample statistic), not a best case, so its population mean is estimable by a sample mean.

The population and empirical leverages and the induced order are

$$F_c = \mathbb{E}_{\mathbf{X} \sim P_{\text{probe}}}[g_c(\mathbf{X})], \quad \hat{F}_c = \frac{1}{n} \sum_{q=1}^n g_c(\mathbf{Z}_q), \quad \hat{\pi} = \text{argsort}_{\downarrow} \hat{F}, \quad \pi^* = \text{argsort}_{\downarrow} F,$$

where F_c is an expectation over the population distribution P_{probe} and $\hat{F}_c$ is the sample mean over the finite set $\mathcal{D}_{\text{probe}}$. All sorting uses a fixed deterministic tie-break. Here $\hat{\pi}$ is the empirical order used by SCS and π^* is the population order used only for analysis. For a target support size $k < C$, the *frontier gap* is $\Delta = F_{\pi_k^*} - F_{\pi_{k+1}^*} \geq 0$, and ν is the sub-Gaussian proxy of $g_c(\mathbf{X})$ for $\mathbf{X} \sim P_{\text{probe}}$ (Assumption 3).

Assumptions for Claim 1. The ranking bound rests on three assumptions. They are stated after the probe objects, leverage function, and expectation are defined, so every symbol in the assumptions has a declared dependency.

Assumption 1 (unbounded scores). *The oracle returns logits in $\mathbb{R}^K$, not probabilities. Hence concentration must use a sub-Gaussian tail, not a bounded-range (Hoeffding) tail.*

Assumption 2 (independent probes). *The finite probe trials $\mathbf{Z}_1, \dots, \mathbf{Z}_n$ in $\mathcal{D}_{\text{probe}}$ are independent draws from P_{probe} ; in practice this is approximated by taking one window per distinct recording.*

Assumption 3 (sub-Gaussian leverage). *For each channel $c \in \mathcal{C}$, the random variable $g_c(\mathbf{X})$ with $\mathbf{X} \sim P_{\text{probe}}$ is ν -sub-Gaussian about its mean F_c . If it is only finite-variance, a Bernstein bound gives the same $1/\Delta^2$ rate with a larger constant.*

4.2 Stage 2: per-trial attack in fixed order

Fix a maximum channel budget $B < C$. For each $k \in \{1, \dots, B\}$, the active channel set is the universal empirical prefix $\mathcal{S}^{(k)} = \{\hat{\pi}_1, \dots, \hat{\pi}_k\}$ ($|\mathcal{S}^{(k)}| = k$), *identical across trials*: only the coefficients $\mathbf{A}_i^{(k)}$ and the realized count k_i^* are per-trial. For trial i at budget k , the perturbation on selected channel c is

$$e_{i,c}^{(k)}(t) = \sum_{m=1}^r a_{i,c,m}^{(k)} b_m(t), \quad c \in \mathcal{S}^{(k)}.$$

The atoms b_m are fixed across trials, but the coefficients $a_{i,c,m}^{(k)}$ are optimized for the particular trial $\mathbf{X}_i$ and selected set size k . Define the trial-and-budget objective

$$J_i^{(k)}(\mathbf{A}) = M(\mathbf{X}_i + \mathbf{E}_{\mathcal{S}^{(k)}, \mathbf{A}}, y_0^i).$$

Stage 2 replaces the random probe coefficients from Stage 1 with optimized trial-specific coefficients. SPSA refines them using two score queries per iteration. At iteration ℓ , with current coefficients $\mathbf{A}_{i,\ell}^{(k)}$, random sign matrix $\mathbf{U}_{i,\ell}^{(k)}$ supported only on rows in $\mathcal{S}^{(k)}$, and radius d_ℓ , the SPSA estimate is

$$\hat{\mathbf{G}}_{i,\ell}^{(k)} = \frac{J_i^{(k)}(\mathbf{A}_{i,\ell}^{(k)} + d_\ell \mathbf{U}_{i,\ell}^{(k)}) - J_i^{(k)}(\mathbf{A}_{i,\ell}^{(k)} - d_\ell \mathbf{U}_{i,\ell}^{(k)})}{2d_\ell} \mathbf{U}_{i,\ell}^{(k)}.$$

The update is

$$\mathbf{A}_{i,\ell+1}^{(k)} = \text{Proj}_{\mathbf{X}_i, \mathcal{S}^{(k)}} \left(\mathbf{A}_{i,\ell}^{(k)} - q_\ell \hat{\mathbf{G}}_{i,\ell}^{(k)} \right),$$

with Spall schedules $d_\ell = d_0/(\ell + 1)^{0.101}$, $q_\ell = q_0/\sqrt{\ell + 1}$, and $\text{Proj}_{\mathbf{X}_i, \mathcal{S}^{(k)}}$ enforcing the peak cap relative to $\mathbf{X}_i$ and zero rows outside $\mathcal{S}^{(k)}$. After the query budget for support size k is spent, let $\mathbf{A}_i^{(k)}$ be the best coefficient matrix found for trial i at that support size.

Metric. The adversarial trial returned for $\mathbf{X}_i$ at budget k is

$$\mathbf{X}_{i,\text{adv}}^{(k)} = \mathbf{X}_i + \mathbf{E}_{\mathcal{S}^{(k)}, \mathbf{A}_i^{(k)}}.$$

The first-success count of trial i is

$$k_i^* = \min\{k \in [B] : M(\mathbf{X}_{i,\text{adv}}^{(k)}, y_0^i) < 0\},$$

undefined if no $k \leq B$ succeeds. The successful set is $\mathcal{I}^* = \{i \in [N] : k_i^* \text{ is defined}\}$. Over this set we report ASR, mean/median of $\{k_i^*\}_{i \in \mathcal{I}^*}$, and the prefix-ASR knees $K90/K95/K100$.

Dependency table for SCS.

Table 1: Index, dependency, shape, and source ledger for SCS notation.

Symbol	Depends on	Size/shape	Source or set
i	none	scalar index	$[N]$
c, c'	none	scalar channel index	$\mathcal{C} = \{1, \dots, C\}$
t	none	scalar time index	$\{0, \dots, T - 1\}$
m	none	scalar atom index	$[r]$
q	none	scalar probe-sample index	$[n]$
j	none	scalar probe-direction index	$[J]$
k	none	scalar support budget	$[B]$, with $B < C$
ℓ	none	scalar optimizer step	SPSA iteration set
$\mathbf{X}_i$	i	$C \times T$ real matrix	$\mathcal{D}$
y_i	i	scalar label	$[K]$
y_0^i	$i, \mathbf{X}_i, f$	scalar predicted label	$[K]$
$\mathcal{D}_{\text{train}}$	HBN passive-task split	set of labeled trials/windows	subset of $\mathcal{D}$
$\mathcal{D}_{\text{val}}$	HBN active-task split	set of labeled trials/windows	subset of $\mathcal{D}$
$\mathcal{D}_{\text{cc}}$	$\mathcal{D}_{\text{val}}, f$	set of clean-correct trials/windows	subset of $\mathcal{D}_{\text{val}}$
$\mathcal{D}_{\text{attack}}$	$\mathcal{D}_{\text{cc}}$, attack sample cap	finite attack-evaluation set	subset of $\mathcal{D}_{\text{cc}}$
f	trained model	map $\mathbb{R}^{C \times T} \rightarrow \mathbb{R}^K$	score oracle
$M(\mathbf{X}, y)$	$\mathbf{X}, y, f$	scalar margin	$\mathbb{R}$
$\mathbf{E}_i$	$i, \mathbf{X}_i, y_0^i, f, r_{\text{max}}$	$C \times T$ real matrix	feasible perturbation set
$s_c(\mathbf{E})$	$\mathbf{E}, c$	scalar indicator	$\{0, 1\}$
$\mathbf{s}(\mathbf{E})$	$\mathbf{E}$	C -vector	$\{0, 1\}^C$
$\mathcal{S}(\mathbf{E})$	$\mathbf{E}$	channel subset	2^C
$\mathcal{F}$	$f_{\text{min}}, f_{\text{max}}, R_f$	set of R_f frequencies	$[f_{\text{min}}, f_{\text{max}}]$
$\mathcal{G}_\varphi$	R_φ	set of R_φ phases	S^1
$\alpha(m), \beta(m)$	$m, \mathcal{M}_{\text{osc}}, \mathcal{F}, \mathcal{G}_\varphi$	pair of scalar grid indices	$[R_f] \times [R_\varphi]$
b_m	m, T, f_s , dictionary grids	length- T unit-norm waveform	$\mathcal{B}$

Symbol	Depends on	Size/shape	Source or set
$\mathbf{A}$	operator argument	$C \times r$ coefficient matrix	one row per channel in $\mathcal{C}$
$\mathbf{E}_{\mathcal{S}, \mathbf{A}}$	$\mathcal{S}, \mathbf{A}, \mathcal{B}$	$C \times T$ perturbation matrix	waveform operator image
P_{probe}	attacker's probe source	distribution over trials	distributions on $\mathbb{R}^{C \times T}$
$\mathbf{Z}_q$	q, P_{probe}	$C \times T$ real matrix	$\mathcal{D}_{\text{probe}}$
$\mathcal{D}_{\text{probe}}$	n, P_{probe}	set of n trials	finite probe sample
$\mathbf{u}_j$	j , probe-bank seed	length- r sign vector	$\{-1, +1\}^r$
$\mathcal{P}$	J , probe-bank seed	set of J sign vectors	probe bank
$\mathbf{E}_{\mathbf{X}, c, j}^{\text{probe}}$	$\mathbf{X}, c, j, \rho, r_{\max}, \mathcal{B}$	$C \times T$ perturbation matrix	capped single-channel probe
$g_c(\mathbf{X})$	$\mathbf{X}, c, \mathcal{P}, \rho, r_{\max}, \mathcal{B}, f$	scalar margin drop	$\mathbb{R}$
F_c	$c, P_{\text{probe}}, \mathcal{P}, \rho, r_{\max}, \mathcal{B}, f$	scalar population leverage	$\mathbb{R}$
$\hat{F}_c$	$c, \mathcal{D}_{\text{probe}}, \mathcal{P}, \rho, r_{\max}, \mathcal{B}, f$	scalar empirical leverage	$\mathbb{R}$
π^*	$F_1, \dots, F_C$	channel permutation	$\mathcal{C}^C$
$\hat{\pi}$	$\hat{F}_1, \dots, \hat{F}_C$	channel permutation	$\mathcal{C}^C$
$\mathcal{S}^{(k)}$	$\hat{\pi}, k$	k -element channel subset	subsets of $\mathcal{C}$
$\mathbf{A}_{i, \ell}^{(k)}$	$i, k, \ell, \mathbf{X}_i, y_0^i, \mathcal{S}^{(k)}$, SPSA seed	$C \times r$ coefficient matrix	SPSA state, zero outside $\mathcal{S}^{(k)}$
$\mathbf{A}_i^{(k)}$	$i, k, \mathbf{X}_i, y_0^i, \mathcal{S}^{(k)}, f$, query budget	$C \times r$ coefficient matrix	best Stage 2 state
$a_{i, c, m}^{(k)}$	$i, c, m, k, \mathbf{A}_i^{(k)}$	scalar coefficient	$\mathbb{R}$
$e_{i, c}^{(k)}(t)$	$i, c, k, t, \mathbf{A}_i^{(k)}, \mathcal{B}$	scalar waveform value	$\mathbb{R}$
$J_i^{(k)}(\mathbf{A})$	$\mathbf{A}, i, k, \mathbf{X}_i, y_0^i, \mathcal{S}^{(k)}, \mathcal{B}, f$	scalar objective value	$\mathbb{R}$
$\mathbf{U}_{i, \ell}^{(k)}$	$i, k, \ell, \mathcal{S}^{(k)}$, SPSA seed	$C \times r$ sign/zero matrix	SPSA perturbation direction
$\hat{\mathbf{G}}_{i, \ell}^{(k)}$	$i, k, \ell, J_i^{(k)}, \mathbf{A}_{i, \ell}^{(k)}, \mathbf{U}_{i, \ell}^{(k)}, d_\ell$	$C \times r$ matrix	SPSA gradient-like estimate
$\mathbf{X}_{i, \text{adv}}^{(k)}$	$i, k, \mathbf{X}_i, \mathcal{S}^{(k)}, \mathbf{A}_i^{(k)}, \mathcal{B}$	$C \times T$ real matrix	attacked trial
k_i^*	$i, B, y_0^i, \{\mathbf{X}_{i, \text{adv}}^{(k)}\}_{k=1}^B, f$	scalar or undefined	$\{1, \dots, B\}$ when successful
$\mathcal{I}^*$	$\{k_i^*\}_{i=1}^N$	trial-index subset	subsets of $[N]$
Δ	F, π^*, k	nonnegative scalar	$\mathbb{R}_{\geq 0}$
ν	$\{g_c(\mathbf{X}) : c \in \mathcal{C}, \mathbf{X} \sim P_{\text{probe}}\}$	nonnegative scalar proxy	sub-Gaussian parameter

For example, $e_{i, c}^{(k)}(t) = \sum_m a_{i, c, m}^{(k)} b_m(t)$ passes because the left side shows the trial, channel, budget, and time dependencies. Writing $e_c(t)$ would fail because it hides the trial and budget dependence.

5 Claim 1 — channel-ranking stability

Theorem 1. *Condition on the fixed dictionary $\mathcal{B}$ and probe bank $\mathcal{P}$. Under Assumptions 2–3, fix $\delta \in (0, 1)$ and a target size $k < C$ with frontier gap $\Delta > 0$. If*

$$n \geq \frac{8\nu^2 \log(2C/\delta)}{\Delta^2}, \quad (7)$$

then with probability at least $1 - \delta$ the empirical top- k set $\{\hat{\pi}_1, \dots, \hat{\pi}_k\}$ equals the population top- k set $\{\pi_1^, \dots, \pi_k^*\}$.*

Lemma 1 (deterministic separation). *If $\max_{c \in \mathcal{C}} |\hat{F}_c - F_c| < \Delta/2$, the empirical and population top- k sets coincide.*

Proof. Let a be any kept channel (population rank $\leq k$) and b any dropped channel (rank $> k$). By definition of Δ , $F_a \geq F_{\pi_k^*} \geq F_{\pi_{k+1}^*} + \Delta \geq F_b + \Delta$. Then $\hat{F}_a > F_a - \Delta/2 \geq F_b + \Delta/2 > \hat{F}_b$. So every kept channel outranks every dropped channel empirically; the top- k set is unchanged. $\square$

Proof of Theorem 1. Each $\hat{F}_c = \frac{1}{n} \sum_{q=1}^n g_c(\mathbf{Z}_q)$ is a mean of n independent (Assumption 2) ν -sub-Gaussian (Assumption 3) terms, hence $\Pr(|\hat{F}_c - F_c| \geq \varepsilon) \leq 2e^{-n\varepsilon^2/(2\nu^2)}$. A union bound over the C channels gives $\Pr(\max_{c \in \mathcal{C}} |\hat{F}_c - F_c| \geq \varepsilon) \leq 2C e^{-n\varepsilon^2/(2\nu^2)}$. Take $\varepsilon = \Delta/2$; the right side is $\leq \delta$ iff $n \geq 8\nu^2 \log(2C/\delta)/\Delta^2$, which is (7). On that event Lemma 1 applies, giving top- k set recovery with probability $\geq 1 - \delta$. $\square$

Insight (the transferable content). The *form* of (7), not the number n , is the result: probe cost grows as $\log C$ in the number of electrodes but as $1/\Delta^2$ in the frontier gap. Closely-ranked electrodes, not many electrodes, cost queries. A small k_i^* is not implied by the ranking theorem alone; it is evidence that the early high- F_c electrodes are sufficient to flip many decisions once the order is stable. We therefore treat (7) as a prediction: estimate empirical proxies for ν and Δ from probe data, then check that ranking agreement saturates near the predicted n^* .

Status (internal)

HBN sweep table/figures present but numbers are from the OLD per-trial greedy; regenerate under the population attack. Claim 1 confirmation (sweep n ; ranking agreement vs n ; overlay n^* from empirical proxies for ν, Δ ; error bars) not yet run. BNCI reference comparison done. Open: whether to headline the law in the abstract; build the `notation.tex` `\newcommand` ledger mirroring §2.